

- 8 (a) In the link reaction, a two-carbon acetyl group is produced from pyruvate. The acetyl group is transferred to coenzyme A to form acetyl coenzyme A (acetyl-coA).

State the terms used to summarise the **two** chemical changes that occur in the link reaction to produce an acetyl group from pyruvate.

.....
..... [2]

- (b) Acetyl-coA combines with oxaloacetate in the Krebs cycle to form citrate. This reaction is catalysed by the enzyme citrate synthase.

Acetyl-coA has a similar shape to succinyl-coA, one of the compounds made in a later part of the Krebs cycle.

An experiment was carried out to investigate the effect of increasing the concentration of acetyl-coA on the activity of citrate synthase. The experiment was repeated, this time adding a solution of succinyl-coA.

Fig. 8.1 shows the results of these experiments.

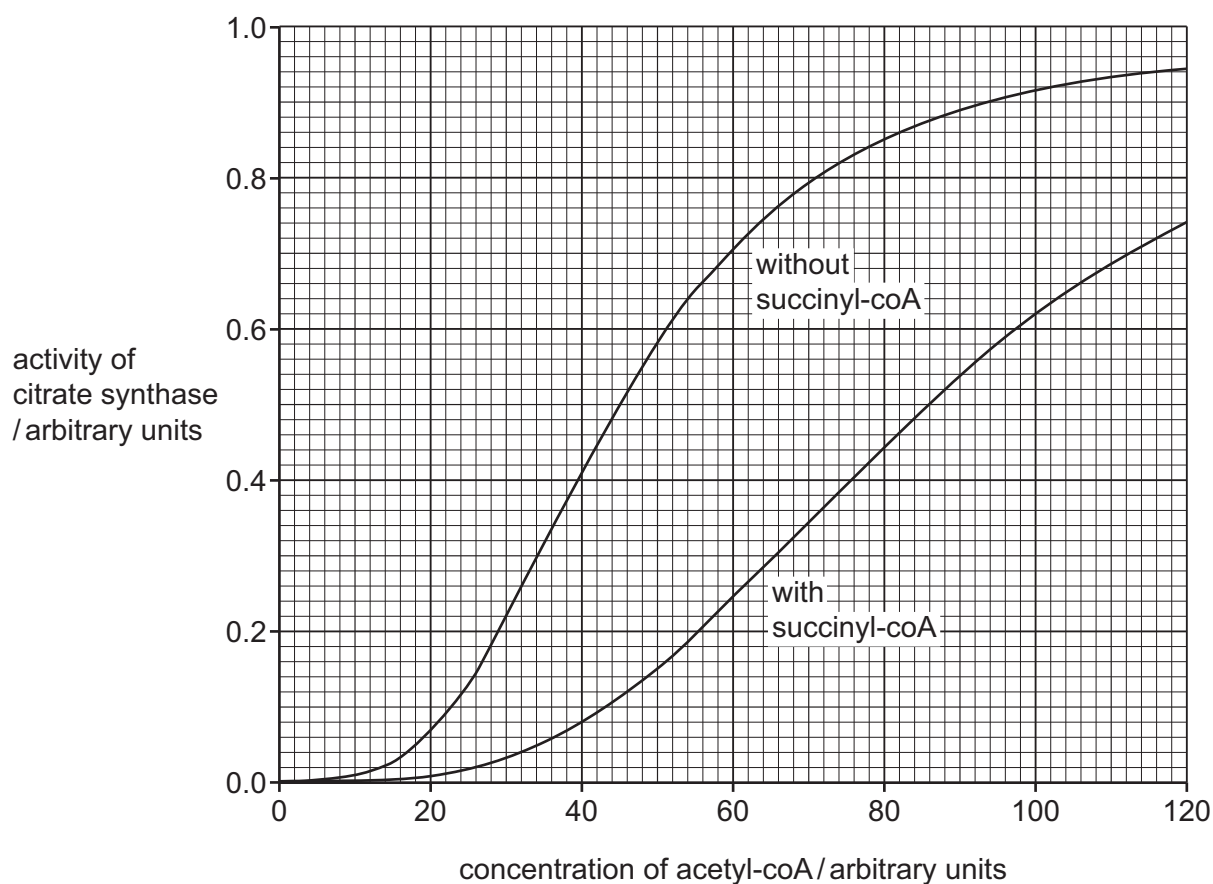


Fig. 8.1



With reference to Fig. 8.1, explain how succinyl-coA could help to regulate the Krebs cycle.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) The coenzyme NAD plays an important role in respiration.

Describe the role of NAD in the stages of aerobic respiration that occur in a mitochondrion.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]